

Continuous Nuclear Reaction-Chain Register from $A = 1$ to $A = 119$ An Experimental Extension of the CPTG Universal Nuclear Reaction Program

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1 Introduction

This document presents an exploratory extension of the Curvature Polarization Transport Gravity (CPTG) universal nuclear reaction program. The program begins with the free-nucleon vertex, proceeds through the deuterium bridge and the mass-three closure pair, and reaches helium-four saturation under the core source, amplitude, rate, transport, and abundance-closure criteria developed for the forthcoming paper on the universal geometric nuclear-reaction theory. That broader theory originated in the commissioned CPTG nuclear-reaction study [1]. The construction is then extended through the represented light- and intermediate-mass reaction network to oxygen-16, followed by the native post-oxygen inventory through silicon-30. A prescribed-trajectory neutron-capture and beta-minus continuation carries the exploratory nuclide graph through mass 119.

Across this progression, one fixed set of scalable CPTG geometric formulas supplies the common coordinate and transport construction. No independent CPTG expression is fitted to each successive mass sector. Beyond the native inventory at $A = 30$, the same geometric organization is applied along the prescribed thermodynamic trajectory, while the individual neutron-capture and beta-minus edge rates are supplied by the selected JINA ReacLib graph through `pynucastro` [3, 4].

The purpose of this extension is to identify, organize, and document the complete set of mass sectors and nuclide states represented in the combined calculation. Every integer mass number in the interval

$$1 \leq A \leq 119 \tag{1}$$

appears exactly once in Table 1. Mass five is included explicitly through the unbound helium-five and lithium-five resonance states relevant to the represented $n + {}^4\text{He}$ and $p + {}^4\text{He}$ channels, although neither state is retained as an abundance-bearing network coordinate [5].

The calculation contains three distinct levels of support. Masses $A = 1\text{--}30$ are included in the native reaction-network inventory. Masses $A = 31\text{--}32$ are obtained from the numerically refined prescribed-trajectory continuation beyond silicon-30. Masses $A = 33\text{--}119$ belong to the larger exploratory neutron-capture and beta-minus graph. These categories preserve the distinction between native-network coverage, the strong first post-silicon diagnostic, and the deeper exploratory continuation.

The mass-120 sector is used only as an absorbing computational boundary. It is therefore excluded from physical interpretation and does not establish an endpoint of the nuclear chain. The present calculation tests how far nonzero reaction flow can be propagated by the fixed geometric framework within the selected network, trajectory, nuclear-data inventory, and numerical resolution.

2 Program Structure and Continuation

The universal light-state sequence is

$$\{n, p\} \longrightarrow {}^2\text{H} \longrightarrow \{{}^3\text{H}, {}^3\text{He}\} \longrightarrow {}^4\text{He}. \quad (2)$$

The absence of a persistent mass-five state does not terminate the reaction network. The represented chain bypasses the mass-five resonances and reaches mass six and higher through multi-body and capture channels. The native basis proceeds through oxygen-16, while the extended native inventory reaches silicon-30 after inclusion of the bridge

$${}^{22}\text{Ne}(n, \gamma){}^{23}\text{Ne} \xrightarrow{\beta^-} {}^{23}\text{Na} + e^- + \bar{\nu}_e. \quad (3)$$

The first post-silicon continuation is

$${}^{30}\text{Si}(n, \gamma){}^{31}\text{Si} \xrightarrow{\beta^-} {}^{31}\text{P} + e^- + \bar{\nu}_e, \quad (4)$$

$${}^{31}\text{P}(n, \gamma){}^{32}\text{P} \xrightarrow{\beta^-} {}^{32}\text{S} + e^- + \bar{\nu}_e. \quad (5)$$

The native trajectory was generated with AlterBBN v2.2 at $\eta_{10} = 6.122$, with the AlterBBN `failsafe` control set to 6 and the species-abundance floor set to 10^{-50} [2]. Under this configuration, inclusion of the bridge gives

$$Y_{23\text{Na}} = 1.0229842 \times 10^{-31}, \quad (6)$$

$$Y_{30\text{Si}} = 1.0741850 \times 10^{-46}. \quad (7)$$

The compact high-resolution post-silicon diagnostic gives

$$Y_{A=31} = 1.2642940 \times 10^{-47}, \quad (8)$$

$$Y_{A=32} = 7.1605139 \times 10^{-48}. \quad (9)$$

These two values come from the compact $A \leq 40$, 32-substep convergence calculation. To keep the continuous register on one numerical basis, Table 1 instead reports the single $A \leq 120$, eight-substep values for every diagnostic mass from $A = 31$ through $A = 119$. The corresponding table values at $A = 31$ and $A = 32$ are $1.2608074 \times 10^{-47}$ and $7.1659590 \times 10^{-48}$, differing from the compact values by approximately 0.276% and 0.076%, respectively.

The prescribed-trajectory calculation was executed with Python 3.13.5 and `pynucastro` 2.11.0. The $A \leq 120$ graph contains 1,079 reachable species including ${}^{30}\text{Si}$ and 2,009 directed reaction edges. The graph retains forward (n, γ) captures and β^- decays only and is integrated by background-interpolated recursive backward Euler on the resulting acyclic graph. The continuous register uses eight integration substeps per native trajectory interval; the compact $A \leq 40$ convergence calculation uses 32 substeps per interval. Reverse photodisintegration and post-silicon charged-particle channels are not included in this prescribed-trajectory graph.

The larger graph carries nonzero trace support through every interior mass from 31 through 119. This rules out interpreting masses 22 and 30 as endpoints of the represented calculation, but it does not establish a finite physical endpoint or native-network coverage beyond silicon-30.

3 Continuous Nuclide Register from Mass 1 Through Mass 119

For each abundance-bearing nuclide i , Y_i denotes its endpoint number abundance per baryon. The abundance assigned to mass sector A is

$$Y_A = \sum_{i:A_i=A} Y_i. \quad (10)$$

Table 1 contains 119 consecutive mass rows and 1,111 listed nuclide or resonance entries. At fixed mass number, entries belonging to different elements are isobars rather than isotopes; the table therefore uses the more precise term *nuclide*, consistent with IUPAC terminology [6]. Masses 1 through 30 reproduce the complete controlling native endpoint inventory. The ^8Be value in that source is the declared 10^{-50} abundance-floor entry and should not be interpreted as a persistent primordial abundance. Masses 31 through 119 reproduce every node in the one-way prescribed-trajectory neutron-capture and beta-minus graph. A listed diagnostic node records rate-library graph membership; it does not, by itself, establish experimental observation or measurable primordial production. Three listed graph nodes— ^{43}Si , ^{47}Si , and ^{51}Si —have exactly zero endpoint abundance in the eight-substep output; every mass-sector sum from $A = 31$ through $A = 119$ remains positive.

The abundance column uses the native endpoint sums for $A = 1$ –30 and the single $A \leq 120$, eight-substep diagnostic output for $A = 31$ –119. It is marked not applicable at $A = 5$ because the listed entries are resonances rather than abundance coordinates. Numerical values are reproduced at the precision stored in the controlling source files; variation in the number of printed trailing digits reflects source-file formatting rather than a change in evidentiary class or an independently assigned uncertainty. The evidence-class codes are defined immediately before the register.

Class codes: **N0**, native universal light-state core; **N1**, native oxygen-16 basis; **N2**, native post-oxygen extension through silicon-30; **D1**, numerically refined prescribed-trajectory continuation; **D2**, exploratory prescribed-trajectory graph; **R**, listed resonance entries without an abundance coordinate.

Table 1: Gap-free register of the nuclides and resonances represented in the combined calculations for every mass number from $A = 1$ through $A = 119$. The column N gives the number of listed entries at each mass.

A	N	Represented nuclides	Y_A	Class
1	2	n , $^1\text{H}(p)$	7.522677×10^{-1}	N0
2	1	$^2\text{H}(\text{D})$	1.860436×10^{-5}	N0
3	2	$^3\text{H}(\text{T})$, ^3He	7.789007×10^{-6}	N0
4	1	^4He	6.191793×10^{-2}	N0
5	2	^5He , ^5Li	not applicable	R
6	1	^6Li	8.214136×10^{-15}	N1
7	2	^7Li , ^7Be	4.024159×10^{-10}	N1

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A	N	Represented nuclides	Y_A	Class
8	3	${}^8\text{Li}$, ${}^8\text{Be}$, ${}^8\text{B}$	2.246811×10^{-34}	N1
9	1	${}^9\text{Be}$	1.86585×10^{-19}	N1
10	2	${}^{10}\text{Be}$, ${}^{10}\text{B}$	2.468639×10^{-21}	N1
11	2	${}^{11}\text{B}$, ${}^{11}\text{C}$	3.495781×10^{-16}	N1
12	3	${}^{12}\text{B}$, ${}^{12}\text{C}$, ${}^{12}\text{N}$	3.582536×10^{-16}	N1
13	2	${}^{13}\text{C}$, ${}^{13}\text{N}$	7.450479×10^{-17}	N1
14	3	${}^{14}\text{C}$, ${}^{14}\text{N}$, ${}^{14}\text{O}$	4.090225×10^{-17}	N1
15	2	${}^{15}\text{N}$, ${}^{15}\text{O}$	9.790313×10^{-21}	N1
16	1	${}^{16}\text{O}$	3.57075×10^{-20}	N1
17	1	${}^{17}\text{O}$	3.134831×10^{-22}	N2
18	2	${}^{18}\text{O}$, ${}^{18}\text{F}$	1.020704×10^{-24}	N2
19	1	${}^{19}\text{F}$	2.790054×10^{-27}	N2
20	1	${}^{20}\text{Ne}$	1.298913×10^{-28}	N2
21	1	${}^{21}\text{Ne}$	5.006889×10^{-30}	N2
22	2	${}^{22}\text{Ne}$, ${}^{22}\text{Na}$	6.868353×10^{-30}	N2
23	2	${}^{23}\text{Ne}$, ${}^{23}\text{Na}$	1.022984×10^{-31}	N2
24	1	${}^{24}\text{Mg}$	2.630659×10^{-35}	N2
25	1	${}^{25}\text{Mg}$	1.678778×10^{-36}	N2
26	2	${}^{26}\text{Mg}$, ${}^{26}\text{Al}$	2.364879×10^{-36}	N2
27	1	${}^{27}\text{Al}$	5.198167×10^{-39}	N2
28	1	${}^{28}\text{Si}$	1.14832×10^{-44}	N2
29	1	${}^{29}\text{Si}$	3.693064×10^{-46}	N2
30	1	${}^{30}\text{Si}$	1.074185×10^{-46}	N2
31	2	${}^{31}\text{Si}$, ${}^{31}\text{P}$	1.260807×10^{-47}	D1

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A	N	Represented nuclides	Y_A	Class
32	3	^{32}Si , ^{32}P , ^{32}S	7.165959×10^{-48}	D1
33	3	^{33}Si , ^{33}P , ^{33}S	6.418241×10^{-49}	D2
34	3	^{34}Si , ^{34}P , ^{34}S	2.919824×10^{-49}	D2
35	4	^{35}Si , ^{35}P , ^{35}S , ^{35}Cl	1.435181×10^{-49}	D2
36	5	^{36}Si , ^{36}P , ^{36}S , ^{36}Cl , ^{36}Ar	5.546139×10^{-50}	D2
37	5	^{37}Si , ^{37}P , ^{37}S , ^{37}Cl , ^{37}Ar	5.235818×10^{-51}	D2
38	5	^{38}Si , ^{38}P , ^{38}S , ^{38}Cl , ^{38}Ar	2.698372×10^{-51}	D2
39	6	^{39}Si , ^{39}P , ^{39}S , ^{39}Cl , ^{39}Ar , ^{39}K	4.871949×10^{-52}	D2
40	7	^{40}Si , ^{40}P , ^{40}S , ^{40}Cl , ^{40}Ar , ^{40}K , ^{40}Ca	1.687562×10^{-52}	D2
41	7	^{41}Si , ^{41}P , ^{41}S , ^{41}Cl , ^{41}Ar , ^{41}K , ^{41}Ca	2.217256×10^{-52}	D2
42	7	^{42}Si , ^{42}P , ^{42}S , ^{42}Cl , ^{42}Ar , ^{42}K , ^{42}Ca	1.8769×10^{-52}	D2
43	7	^{43}Si , ^{43}P , ^{43}S , ^{43}Cl , ^{43}Ar , ^{43}K , ^{43}Ca	8.938921×10^{-53}	D2
44	7	^{44}Si , ^{44}P , ^{44}S , ^{44}Cl , ^{44}Ar , ^{44}K , ^{44}Ca	1.271115×10^{-52}	D2
45	8	^{45}Si , ^{45}P , ^{45}S , ^{45}Cl , ^{45}Ar , ^{45}K , ^{45}Ca , ^{45}Sc	5.279289×10^{-53}	D2
46	9	^{46}Si , ^{46}P , ^{46}S , ^{46}Cl , ^{46}Ar , ^{46}K , ^{46}Ca , ^{46}Sc , ^{46}Ti	4.728164×10^{-53}	D2
47	9	^{47}Si , ^{47}P , ^{47}S , ^{47}Cl , ^{47}Ar , ^{47}K , ^{47}Ca , ^{47}Sc , ^{47}Ti	3.905347×10^{-53}	D2
48	9	^{48}Si , ^{48}P , ^{48}S , ^{48}Cl , ^{48}Ar , ^{48}K , ^{48}Ca , ^{48}Sc , ^{48}Ti	3.351341×10^{-53}	D2
49	9	^{49}Si , ^{49}P , ^{49}S , ^{49}Cl , ^{49}Ar , ^{49}K , ^{49}Ca , ^{49}Sc , ^{49}Ti	2.322291×10^{-53}	D2
50	9	^{50}Si , ^{50}P , ^{50}S , ^{50}Cl , ^{50}Ar , ^{50}K , ^{50}Ca , ^{50}Sc , ^{50}Ti	1.119949×10^{-53}	D2
51	10	^{51}Si , ^{51}P , ^{51}S , ^{51}Cl , ^{51}Ar , ^{51}K , ^{51}Ca , ^{51}Sc , ^{51}Ti , ^{51}V	6.471204×10^{-54}	D2
52	9	^{52}S , ^{52}Cl , ^{52}Ar , ^{52}K , ^{52}Ca , ^{52}Sc , ^{52}Ti , ^{52}V , ^{52}Cr	4.588623×10^{-54}	D2
53	9	^{53}S , ^{53}Cl , ^{53}Ar , ^{53}K , ^{53}Ca , ^{53}Sc , ^{53}Ti , ^{53}V , ^{53}Cr	3.067894×10^{-54}	D2
54	9	^{54}S , ^{54}Cl , ^{54}Ar , ^{54}K , ^{54}Ca , ^{54}Sc , ^{54}Ti , ^{54}V , ^{54}Cr	1.037553×10^{-54}	D2
55	10	^{55}S , ^{55}Cl , ^{55}Ar , ^{55}K , ^{55}Ca , ^{55}Sc , ^{55}Ti , ^{55}V , ^{55}Cr , ^{55}Mn	1.169812×10^{-54}	D2

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A	N	Represented nuclides	Y_A	Class
56	11	^{56}S , ^{56}Cl , ^{56}Ar , ^{56}K , ^{56}Ca , ^{56}Sc , ^{56}Ti , ^{56}V , ^{56}Cr , ^{56}Mn , ^{56}Fe	1.765245×10^{-54}	D2
57	11	^{57}S , ^{57}Cl , ^{57}Ar , ^{57}K , ^{57}Ca , ^{57}Sc , ^{57}Ti , ^{57}V , ^{57}Cr , ^{57}Mn , ^{57}Fe	7.957791×10^{-55}	D2
58	11	^{58}S , ^{58}Cl , ^{58}Ar , ^{58}K , ^{58}Ca , ^{58}Sc , ^{58}Ti , ^{58}V , ^{58}Cr , ^{58}Mn , ^{58}Fe	5.671194×10^{-55}	D2
59	11	^{59}Cl , ^{59}Ar , ^{59}K , ^{59}Ca , ^{59}Sc , ^{59}Ti , ^{59}V , ^{59}Cr , ^{59}Mn , ^{59}Fe , ^{59}Co	4.114257×10^{-55}	D2
60	12	^{60}Cl , ^{60}Ar , ^{60}K , ^{60}Ca , ^{60}Sc , ^{60}Ti , ^{60}V , ^{60}Cr , ^{60}Mn , ^{60}Fe , ^{60}Co , ^{60}Ni	9.994849×10^{-55}	D2
61	12	^{61}Cl , ^{61}Ar , ^{61}K , ^{61}Ca , ^{61}Sc , ^{61}Ti , ^{61}V , ^{61}Cr , ^{61}Mn , ^{61}Fe , ^{61}Co , ^{61}Ni	4.205458×10^{-55}	D2
62	11	^{62}Ar , ^{62}K , ^{62}Ca , ^{62}Sc , ^{62}Ti , ^{62}V , ^{62}Cr , ^{62}Mn , ^{62}Fe , ^{62}Co , ^{62}Ni	6.831138×10^{-55}	D2
63	12	^{63}Ar , ^{63}K , ^{63}Ca , ^{63}Sc , ^{63}Ti , ^{63}V , ^{63}Cr , ^{63}Mn , ^{63}Fe , ^{63}Co , ^{63}Ni , ^{63}Cu	3.540761×10^{-55}	D2
64	13	^{64}Ar , ^{64}K , ^{64}Ca , ^{64}Sc , ^{64}Ti , ^{64}V , ^{64}Cr , ^{64}Mn , ^{64}Fe , ^{64}Co , ^{64}Ni , ^{64}Cu , ^{64}Zn	3.755918×10^{-55}	D2
65	12	^{65}K , ^{65}Ca , ^{65}Sc , ^{65}Ti , ^{65}V , ^{65}Cr , ^{65}Mn , ^{65}Fe , ^{65}Co , ^{65}Ni , ^{65}Cu , ^{65}Zn	1.504781×10^{-55}	D2
66	12	^{66}K , ^{66}Ca , ^{66}Sc , ^{66}Ti , ^{66}V , ^{66}Cr , ^{66}Mn , ^{66}Fe , ^{66}Co , ^{66}Ni , ^{66}Cu , ^{66}Zn	4.948077×10^{-55}	D2
67	12	^{67}K , ^{67}Ca , ^{67}Sc , ^{67}Ti , ^{67}V , ^{67}Cr , ^{67}Mn , ^{67}Fe , ^{67}Co , ^{67}Ni , ^{67}Cu , ^{67}Zn	1.517976×10^{-55}	D2
68	11	^{68}Ca , ^{68}Sc , ^{68}Ti , ^{68}V , ^{68}Cr , ^{68}Mn , ^{68}Fe , ^{68}Co , ^{68}Ni , ^{68}Cu , ^{68}Zn	3.464737×10^{-55}	D2
69	12	^{69}Ca , ^{69}Sc , ^{69}Ti , ^{69}V , ^{69}Cr , ^{69}Mn , ^{69}Fe , ^{69}Co , ^{69}Ni , ^{69}Cu , ^{69}Zn , ^{69}Ga	2.851959×10^{-55}	D2
70	13	^{70}Ca , ^{70}Sc , ^{70}Ti , ^{70}V , ^{70}Cr , ^{70}Mn , ^{70}Fe , ^{70}Co , ^{70}Ni , ^{70}Cu , ^{70}Zn , ^{70}Ga , ^{70}Ge	1.086518×10^{-55}	D2
71	13	^{71}Ca , ^{71}Sc , ^{71}Ti , ^{71}V , ^{71}Cr , ^{71}Mn , ^{71}Fe , ^{71}Co , ^{71}Ni , ^{71}Cu , ^{71}Zn , ^{71}Ga , ^{71}Ge	1.460832×10^{-55}	D2
72	12	^{72}Sc , ^{72}Ti , ^{72}V , ^{72}Cr , ^{72}Mn , ^{72}Fe , ^{72}Co , ^{72}Ni , ^{72}Cu , ^{72}Zn , ^{72}Ga , ^{72}Ge	1.863515×10^{-55}	D2
73	12	^{73}Sc , ^{73}Ti , ^{73}V , ^{73}Cr , ^{73}Mn , ^{73}Fe , ^{73}Co , ^{73}Ni , ^{73}Cu , ^{73}Zn , ^{73}Ga , ^{73}Ge	8.537773×10^{-56}	D2
74	12	^{74}Sc , ^{74}Ti , ^{74}V , ^{74}Cr , ^{74}Mn , ^{74}Fe , ^{74}Co , ^{74}Ni , ^{74}Cu , ^{74}Zn , ^{74}Ga , ^{74}Ge	2.819861×10^{-55}	D2
75	12	^{75}Ti , ^{75}V , ^{75}Cr , ^{75}Mn , ^{75}Fe , ^{75}Co , ^{75}Ni , ^{75}Cu , ^{75}Zn , ^{75}Ga , ^{75}Ge , ^{75}As	1.851494×10^{-55}	D2
76	13	^{76}Ti , ^{76}V , ^{76}Cr , ^{76}Mn , ^{76}Fe , ^{76}Co , ^{76}Ni , ^{76}Cu , ^{76}Zn , ^{76}Ga , ^{76}Ge , ^{76}As , ^{76}Se	5.175386×10^{-56}	D2
77	13	^{77}Ti , ^{77}V , ^{77}Cr , ^{77}Mn , ^{77}Fe , ^{77}Co , ^{77}Ni , ^{77}Cu , ^{77}Zn , ^{77}Ga , ^{77}Ge , ^{77}As , ^{77}Se	1.020116×10^{-55}	D2
78	12	^{78}V , ^{78}Cr , ^{78}Mn , ^{78}Fe , ^{78}Co , ^{78}Ni , ^{78}Cu , ^{78}Zn , ^{78}Ga , ^{78}Ge , ^{78}As , ^{78}Se	2.107071×10^{-55}	D2
79	13	^{79}V , ^{79}Cr , ^{79}Mn , ^{79}Fe , ^{79}Co , ^{79}Ni , ^{79}Cu , ^{79}Zn , ^{79}Ga , ^{79}Ge , ^{79}As , ^{79}Se , ^{79}Br	5.182884×10^{-56}	D2

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A	N	Represented nuclides	Y_A	Class
80	14	^{80}V , ^{80}Cr , ^{80}Mn , ^{80}Fe , ^{80}Co , ^{80}Ni , ^{80}Cu , ^{80}Zn , ^{80}Ga , ^{80}Ge , ^{80}As , ^{80}Se , ^{80}Br , ^{80}Kr	1.016253×10^{-55}	D2
81	13	^{81}Cr , ^{81}Mn , ^{81}Fe , ^{81}Co , ^{81}Ni , ^{81}Cu , ^{81}Zn , ^{81}Ga , ^{81}Ge , ^{81}As , ^{81}Se , ^{81}Br , ^{81}Kr	1.210256×10^{-55}	D2
82	13	^{82}Cr , ^{82}Mn , ^{82}Fe , ^{82}Co , ^{82}Ni , ^{82}Cu , ^{82}Zn , ^{82}Ga , ^{82}Ge , ^{82}As , ^{82}Se , ^{82}Br , ^{82}Kr	1.046129×10^{-55}	D2
83	13	^{83}Cr , ^{83}Mn , ^{83}Fe , ^{83}Co , ^{83}Ni , ^{83}Cu , ^{83}Zn , ^{83}Ga , ^{83}Ge , ^{83}As , ^{83}Se , ^{83}Br , ^{83}Kr	1.597622×10^{-55}	D2
84	13	^{84}Cr , ^{84}Mn , ^{84}Fe , ^{84}Co , ^{84}Ni , ^{84}Cu , ^{84}Zn , ^{84}Ga , ^{84}Ge , ^{84}As , ^{84}Se , ^{84}Br , ^{84}Kr	2.743322×10^{-55}	D2
85	13	^{85}Mn , ^{85}Fe , ^{85}Co , ^{85}Ni , ^{85}Cu , ^{85}Zn , ^{85}Ga , ^{85}Ge , ^{85}As , ^{85}Se , ^{85}Br , ^{85}Kr , ^{85}Rb	2.316254×10^{-55}	D2
86	14	^{86}Mn , ^{86}Fe , ^{86}Co , ^{86}Ni , ^{86}Cu , ^{86}Zn , ^{86}Ga , ^{86}Ge , ^{86}As , ^{86}Se , ^{86}Br , ^{86}Kr , ^{86}Rb , ^{86}Sr	1.44283×10^{-55}	D2
87	14	^{87}Mn , ^{87}Fe , ^{87}Co , ^{87}Ni , ^{87}Cu , ^{87}Zn , ^{87}Ga , ^{87}Ge , ^{87}As , ^{87}Se , ^{87}Br , ^{87}Kr , ^{87}Rb , ^{87}Sr	6.053177×10^{-56}	D2
88	13	^{88}Fe , ^{88}Co , ^{88}Ni , ^{88}Cu , ^{88}Zn , ^{88}Ga , ^{88}Ge , ^{88}As , ^{88}Se , ^{88}Br , ^{88}Kr , ^{88}Rb , ^{88}Sr	2.926046×10^{-56}	D2
89	14	^{89}Fe , ^{89}Co , ^{89}Ni , ^{89}Cu , ^{89}Zn , ^{89}Ga , ^{89}Ge , ^{89}As , ^{89}Se , ^{89}Br , ^{89}Kr , ^{89}Rb , ^{89}Sr , ^{89}Y	6.1468×10^{-56}	D2
90	15	^{90}Fe , ^{90}Co , ^{90}Ni , ^{90}Cu , ^{90}Zn , ^{90}Ga , ^{90}Ge , ^{90}As , ^{90}Se , ^{90}Br , ^{90}Kr , ^{90}Rb , ^{90}Sr , ^{90}Y , ^{90}Zr	4.745019×10^{-56}	D2
91	14	^{91}Co , ^{91}Ni , ^{91}Cu , ^{91}Zn , ^{91}Ga , ^{91}Ge , ^{91}As , ^{91}Se , ^{91}Br , ^{91}Kr , ^{91}Rb , ^{91}Sr , ^{91}Y , ^{91}Zr	4.245786×10^{-56}	D2
92	14	^{92}Co , ^{92}Ni , ^{92}Cu , ^{92}Zn , ^{92}Ga , ^{92}Ge , ^{92}As , ^{92}Se , ^{92}Br , ^{92}Kr , ^{92}Rb , ^{92}Sr , ^{92}Y , ^{92}Zr	3.309035×10^{-56}	D2
93	15	^{93}Co , ^{93}Ni , ^{93}Cu , ^{93}Zn , ^{93}Ga , ^{93}Ge , ^{93}As , ^{93}Se , ^{93}Br , ^{93}Kr , ^{93}Rb , ^{93}Sr , ^{93}Y , ^{93}Zr , ^{93}Nb	2.900263×10^{-56}	D2
94	15	^{94}Ni , ^{94}Cu , ^{94}Zn , ^{94}Ga , ^{94}Ge , ^{94}As , ^{94}Se , ^{94}Br , ^{94}Kr , ^{94}Rb , ^{94}Sr , ^{94}Y , ^{94}Zr , ^{94}Nb , ^{94}Mo	3.931671×10^{-56}	D2
95	15	^{95}Ni , ^{95}Cu , ^{95}Zn , ^{95}Ga , ^{95}Ge , ^{95}As , ^{95}Se , ^{95}Br , ^{95}Kr , ^{95}Rb , ^{95}Sr , ^{95}Y , ^{95}Zr , ^{95}Nb , ^{95}Mo	2.978948×10^{-56}	D2
96	15	^{96}Ni , ^{96}Cu , ^{96}Zn , ^{96}Ga , ^{96}Ge , ^{96}As , ^{96}Se , ^{96}Br , ^{96}Kr , ^{96}Rb , ^{96}Sr , ^{96}Y , ^{96}Zr , ^{96}Nb , ^{96}Mo	1.634314×10^{-56}	D2
97	14	^{97}Cu , ^{97}Zn , ^{97}Ga , ^{97}Ge , ^{97}As , ^{97}Se , ^{97}Br , ^{97}Kr , ^{97}Rb , ^{97}Sr , ^{97}Y , ^{97}Zr , ^{97}Nb , ^{97}Mo	2.386422×10^{-56}	D2
98	14	^{98}Cu , ^{98}Zn , ^{98}Ga , ^{98}Ge , ^{98}As , ^{98}Se , ^{98}Br , ^{98}Kr , ^{98}Rb , ^{98}Sr , ^{98}Y , ^{98}Zr , ^{98}Nb , ^{98}Mo	1.19575×10^{-56}	D2

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A	N	Represented nuclides	Y_A	Class
99	16	^{99}Cu , ^{99}Zn , ^{99}Ga , ^{99}Ge , ^{99}As , ^{99}Se , ^{99}Br , ^{99}Kr , ^{99}Rb , ^{99}Sr , ^{99}Y , ^{99}Zr , ^{99}Nb , ^{99}Mo , ^{99}Tc , ^{99}Ru	4.010949×10^{-57}	D2
100	16	^{100}Cu , ^{100}Zn , ^{100}Ga , ^{100}Ge , ^{100}As , ^{100}Se , ^{100}Br , ^{100}Kr , ^{100}Rb , ^{100}Sr , ^{100}Y , ^{100}Zr , ^{100}Nb , ^{100}Mo , ^{100}Tc , ^{100}Ru	4.66066×10^{-57}	D2
101	15	^{101}Zn , ^{101}Ga , ^{101}Ge , ^{101}As , ^{101}Se , ^{101}Br , ^{101}Kr , ^{101}Rb , ^{101}Sr , ^{101}Y , ^{101}Zr , ^{101}Nb , ^{101}Mo , ^{101}Tc , ^{101}Ru	3.713053×10^{-57}	D2
102	15	^{102}Zn , ^{102}Ga , ^{102}Ge , ^{102}As , ^{102}Se , ^{102}Br , ^{102}Kr , ^{102}Rb , ^{102}Sr , ^{102}Y , ^{102}Zr , ^{102}Nb , ^{102}Mo , ^{102}Tc , ^{102}Ru	8.071407×10^{-57}	D2
103	16	^{103}Zn , ^{103}Ga , ^{103}Ge , ^{103}As , ^{103}Se , ^{103}Br , ^{103}Kr , ^{103}Rb , ^{103}Sr , ^{103}Y , ^{103}Zr , ^{103}Nb , ^{103}Mo , ^{103}Tc , ^{103}Ru , ^{103}Rh	2.487525×10^{-57}	D2
104	16	^{104}Ga , ^{104}Ge , ^{104}As , ^{104}Se , ^{104}Br , ^{104}Kr , ^{104}Rb , ^{104}Sr , ^{104}Y , ^{104}Zr , ^{104}Nb , ^{104}Mo , ^{104}Tc , ^{104}Ru , ^{104}Rh , ^{104}Pd	6.492919×10^{-57}	D2
105	16	^{105}Ga , ^{105}Ge , ^{105}As , ^{105}Se , ^{105}Br , ^{105}Kr , ^{105}Rb , ^{105}Sr , ^{105}Y , ^{105}Zr , ^{105}Nb , ^{105}Mo , ^{105}Tc , ^{105}Ru , ^{105}Rh , ^{105}Pd	8.021328×10^{-57}	D2
106	16	^{106}Ga , ^{106}Ge , ^{106}As , ^{106}Se , ^{106}Br , ^{106}Kr , ^{106}Rb , ^{106}Sr , ^{106}Y , ^{106}Zr , ^{106}Nb , ^{106}Mo , ^{106}Tc , ^{106}Ru , ^{106}Rh , ^{106}Pd	3.052984×10^{-57}	D2
107	16	^{107}Ge , ^{107}As , ^{107}Se , ^{107}Br , ^{107}Kr , ^{107}Rb , ^{107}Sr , ^{107}Y , ^{107}Zr , ^{107}Nb , ^{107}Mo , ^{107}Tc , ^{107}Ru , ^{107}Rh , ^{107}Pd , ^{107}Ag	4.27056×10^{-57}	D2
108	17	^{108}Ge , ^{108}As , ^{108}Se , ^{108}Br , ^{108}Kr , ^{108}Rb , ^{108}Sr , ^{108}Y , ^{108}Zr , ^{108}Nb , ^{108}Mo , ^{108}Tc , ^{108}Ru , ^{108}Rh , ^{108}Pd , ^{108}Ag , ^{108}Cd	6.845493×10^{-57}	D2
109	17	^{109}Ge , ^{109}As , ^{109}Se , ^{109}Br , ^{109}Kr , ^{109}Rb , ^{109}Sr , ^{109}Y , ^{109}Zr , ^{109}Nb , ^{109}Mo , ^{109}Tc , ^{109}Ru , ^{109}Rh , ^{109}Pd , ^{109}Ag , ^{109}Cd	2.022383×10^{-57}	D2
110	16	^{110}As , ^{110}Se , ^{110}Br , ^{110}Kr , ^{110}Rb , ^{110}Sr , ^{110}Y , ^{110}Zr , ^{110}Nb , ^{110}Mo , ^{110}Tc , ^{110}Ru , ^{110}Rh , ^{110}Pd , ^{110}Ag , ^{110}Cd	5.998031×10^{-57}	D2
111	16	^{111}As , ^{111}Se , ^{111}Br , ^{111}Kr , ^{111}Rb , ^{111}Sr , ^{111}Y , ^{111}Zr , ^{111}Nb , ^{111}Mo , ^{111}Tc , ^{111}Ru , ^{111}Rh , ^{111}Pd , ^{111}Ag , ^{111}Cd	2.259657×10^{-57}	D2
112	16	^{112}As , ^{112}Se , ^{112}Br , ^{112}Kr , ^{112}Rb , ^{112}Sr , ^{112}Y , ^{112}Zr , ^{112}Nb , ^{112}Mo , ^{112}Tc , ^{112}Ru , ^{112}Rh , ^{112}Pd , ^{112}Ag , ^{112}Cd	6.772206×10^{-57}	D2
113	16	^{113}Se , ^{113}Br , ^{113}Kr , ^{113}Rb , ^{113}Sr , ^{113}Y , ^{113}Zr , ^{113}Nb , ^{113}Mo , ^{113}Tc , ^{113}Ru , ^{113}Rh , ^{113}Pd , ^{113}Ag , ^{113}Cd , ^{113}In	1.479664×10^{-57}	D2
114	17	^{114}Se , ^{114}Br , ^{114}Kr , ^{114}Rb , ^{114}Sr , ^{114}Y , ^{114}Zr , ^{114}Nb , ^{114}Mo , ^{114}Tc , ^{114}Ru , ^{114}Rh , ^{114}Pd , ^{114}Ag , ^{114}Cd , ^{114}In , ^{114}Sn	7.965147×10^{-57}	D2
115	17	^{115}Se , ^{115}Br , ^{115}Kr , ^{115}Rb , ^{115}Sr , ^{115}Y , ^{115}Zr , ^{115}Nb , ^{115}Mo , ^{115}Tc , ^{115}Ru , ^{115}Rh , ^{115}Pd , ^{115}Ag , ^{115}Cd , ^{115}In , ^{115}Sn	3.090037×10^{-57}	D2
116	17	^{116}Se , ^{116}Br , ^{116}Kr , ^{116}Rb , ^{116}Sr , ^{116}Y , ^{116}Zr , ^{116}Nb , ^{116}Mo , ^{116}Tc , ^{116}Ru , ^{116}Rh , ^{116}Pd , ^{116}Ag , ^{116}Cd , ^{116}In , ^{116}Sn	2.0995×10^{-57}	D2

Continued on next page

A	N	Represented nuclides	Y_A	Class
117	16	$^{117}\text{Br}, ^{117}\text{Kr}, ^{117}\text{Rb}, ^{117}\text{Sr}, ^{117}\text{Y}, ^{117}\text{Zr}, ^{117}\text{Nb}, ^{117}\text{Mo}, ^{117}\text{Tc}, ^{117}\text{Ru}, ^{117}\text{Rh},$ $^{117}\text{Pd}, ^{117}\text{Ag}, ^{117}\text{Cd}, ^{117}\text{In}, ^{117}\text{Sn}$	5.006565×10^{-57}	D2
118	16	$^{118}\text{Br}, ^{118}\text{Kr}, ^{118}\text{Rb}, ^{118}\text{Sr}, ^{118}\text{Y}, ^{118}\text{Zr}, ^{118}\text{Nb}, ^{118}\text{Mo}, ^{118}\text{Tc}, ^{118}\text{Ru}, ^{118}\text{Rh},$ $^{118}\text{Pd}, ^{118}\text{Ag}, ^{118}\text{Cd}, ^{118}\text{In}, ^{118}\text{Sn}$	7.760345×10^{-57}	D2
119	16	$^{119}\text{Br}, ^{119}\text{Kr}, ^{119}\text{Rb}, ^{119}\text{Sr}, ^{119}\text{Y}, ^{119}\text{Zr}, ^{119}\text{Nb}, ^{119}\text{Mo}, ^{119}\text{Tc}, ^{119}\text{Ru}, ^{119}\text{Rh},$ $^{119}\text{Pd}, ^{119}\text{Ag}, ^{119}\text{Cd}, ^{119}\text{In}, ^{119}\text{Sn}$	2.873667×10^{-57}	D2

4 The Mass-Five Gap

Mass five is present in the continuous register so that the data contain no missing mass row. The two mass-five resonances explicitly represented in this construction,

$$^5\text{He} \quad \text{and} \quad ^5\text{Li}, \quad (11)$$

are unbound resonances:

$$^5\text{He} \longrightarrow ^4\text{He} + n, \quad (12)$$

$$^5\text{Li} \longrightarrow ^4\text{He} + p. \quad (13)$$

They are broad, particle-unstable resonances and do not accumulate as persistent abundance-bearing states under the conditions represented here. The row is not an exhaustive catalog of every evaluated $A = 5$ system. It records the two resonances relevant to the represented $n + ^4\text{He}$ and $p + ^4\text{He}$ channels and assigns neither an endpoint abundance nor an independent native coordinate. The network can bypass the gap through channels such as

$$^2\text{H} + ^4\text{He} \longrightarrow ^6\text{Li} + \gamma, \quad (14)$$

$$^3\text{H} + ^4\text{He} \longrightarrow ^7\text{Li} + \gamma, \quad (15)$$

$$^3\text{He} + ^4\text{He} \longrightarrow ^7\text{Be} + \gamma. \quad (16)$$

The mass-five row is consequently a physical gap designation, not a missing-data entry.

5 Completeness and Support Classification

The continuous register satisfies the following checks:

$$N_{\text{mass rows}} = 119, \quad (17)$$

$$N_{\text{missing mass rows}} = 0, \quad (18)$$

$$N_{\text{listed nuclide/resonance entries}} = 1111, \quad (19)$$

$$N_{\text{zero endpoint graph nodes}} = 3, \quad (20)$$

$$A_{\text{min}} = 1, \quad (21)$$

$$A_{\text{max}} = 119. \quad (22)$$

The evidence classification is

$$A = 1\text{--}30 : \text{native inventory}, \quad (23)$$

$$A = 31\text{--}32 : \text{strong prescribed-trajectory diagnostic}, \quad (24)$$

$$A = 33\text{--}119 : \text{exploratory prescribed-trajectory graph}. \quad (25)$$

The nuclide lists at masses 31 through 119 describe membership in the reaction graph. They are not claims that all listed nuclides obtain measurable primordial abundances, nor are they a catalog of every experimentally known nuclide at those masses. Some ReacLib graph nodes are supported by evaluated or theoretical rates; graph membership alone is not an experimental-discovery classification. The companion nuclide-level CSV preserves the same register in machine-readable form, including A , nuclide name, proton number, endpoint abundance when applicable, evidence class, and the explicit mass-five resonance note. The three zero-endpoint graph nodes are retained because they are members of the generated reaction graph even though their final floating-point abundance is zero.

6 Scientific Boundary

The complete register demonstrates continuity of the represented calculation but does not erase the distinction between native-network and diagnostic support. Native-network coverage presently ends at silicon-30. The first post-silicon block is numerically strong on the prescribed thermodynamic trajectory. The deeper tail remains exploratory and becomes increasingly sensitive to temporal refinement and network-boundary placement. The mass-120 terminal bin is excluded because it was imposed as an absorbing boundary.

Accordingly, the current result is:

The combined native and prescribed-trajectory calculation contains an explicit, gap-free mass register from $A = 1$ through $A = 119$. Nonzero diagnostic support reaches every interior mass above silicon through $A = 119$, but no finite physical endpoint has yet been identified.

7 Data and Code Availability

The controlling release bundle is:

CPTG_Post016_Si30Bridge_AbsoluteLimitDiagnostic_20260716_r01.zip

with SHA-256:

76ee1c5c637069972c231cfda721afbcae1fb8975ca6a128ac2bea5abf79410a

Within the CPTG repository, the repository-relative release location is:

<https://github.com/CLG2025/CPTG/tree/main/nuclear-reactions>

The principal machine-readable inputs and implementation files are:

- RESULTS/TRACE_DIAGNOSTIC/raw_endpoint.csv
SHA-256 9e234e2d8f672ad77fbfa1faa26c7fe20020e62325f07eabaa2ae5fdae1272e7

- RESULTS/TRACE_DIAGNOSTIC/SI30_NEUTRON_BETA_A120_S8_TRI_ENDPOINT.csv
SHA-256 aecc87d9a58b5e16f2977c2857ddcd4045fd56cf5172ea397b90d1808cda8180
- RESULTS/TRACE_DIAGNOSTIC/SI30_NEUTRON_BETA_A120_S8_TRI_BY_MASS.csv
SHA-256 a96b8914c977766e981e017a0054c40b52935a6fa5ab1a2aaf04a3567a797817
- SOURCE/si30_neutron_beta_triangular_fast.py
SHA-256 9adb230db299020e7be09fc4e5308a8af70814939242fd9b762e50ec0f73771c

The upstream release identities retained in the bundle are:

- CPTG_Post016_TheoryExtension_AI_Handoff_20260716_r01.zip
SHA-256 20cbaca5676f9d06dd1cea2be6b45c9c1ecd60dfc07b75032cdc1221b2330c4a
- CPTG_Post016_RobustFrontierConfirmation_20260716_r02.zip
SHA-256 75a1855ebf7e3460095ace4383d9c704d161a5f5a7a01cc75f1184f6a3688662
- CPTG_Ne23_BBN_vs_ExplosiveRegime_Discrimination_20260716_r01.zip
SHA-256 8757fa29971af6f6375978ee6731b6d04c425e0916bfafc5c6a4de512ef10b60

The native trajectory was produced with AlterBBN v2.2 at $\eta_{10} = 6.122$, `failsafe`=6, and abundance floor 10^{-50} . The prescribed-trajectory continuation used Python 3.13.5, `pynucastro` 2.11.0, a maximum network mass of $A = 120$, and background-interpolated recursive backward Euler. The terminal $A = 120$ sector is an imposed absorbing boundary and is excluded from the physical register and endpoint interpretation.

References

- [1] Carter L Glass Jr, *Geometric Nuclear Reaction Theory in CPTG, Deuterium-Proton Capture and Primordial Mass-Seven Transport*, 2026, DOI: 10.13140/RG.2.2.26426.35521.
- [2] A. Arbey, J. Auffinger, K. P. Hickerson, and E. S. Jenssen, “AlterBBN v2: A public code for calculating Big-Bang nucleosynthesis constraints in alternative cosmologies,” *Computer Physics Communications* **248**, 106982 (2020), DOI: 10.1016/j.cpc.2019.106982.
- [3] R. H. Cyburt et al., “The JINA REACLIB Database: Its Recent Updates and Impact on Type-I X-Ray Bursts,” *The Astrophysical Journal Supplement Series* **189**, 240–252 (2010), DOI: 10.1088/0067-0049/189/1/240.
- [4] A. I. Smith et al., “pynucastro: A Python Library for Nuclear Astrophysics,” *The Astrophysical Journal* **947**, 65 (2023), DOI: 10.3847/1538-4357/acbaff.
- [5] D. R. Tilley, C. M. Cheves, J. L. Godwin, G. M. Hale, H. M. Hofmann, J. H. Kelley, C. G. Sheu, and H. R. Weller, “Energy Levels of Light Nuclei $A = 5, 6, 7$,” *Nuclear Physics A* **708**, 3–163 (2002), DOI: 10.1016/S0375-9474(02)00597-3.
- [6] International Union of Pure and Applied Chemistry, *Compendium of Chemical Terminology (the Gold Book)*, 5th ed., online version 5.0.0 (2025): “nuclide,” DOI 10.1351/goldbook.N04257; “isotopes,” DOI 10.1351/goldbook.I03331; and “isobars,” DOI 10.1351/goldbook.I03263.